

Zachary Benson

Java Engineer

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PROFESSIONAL SUMMARY

As a software engineer with 8 years of experience, I bring strong technical expertise, a collaborative mindset, and a deep commitment to building high-quality software. I'm driven by curiosity and a constant desire to improve system performance, code quality, and overall engineering excellence. I'm a highly motivated, results-oriented engineer who focuses on delivering meaningful outcomes. While I'm energized by achieving the final goal, I approach the process with discipline, adaptability, and a willingness to learn. I thrive in team environments that value open communication and shared ownership, yet I'm equally effective working independently with accountability and focus. With a customer-centric perspective, I prioritize solving high-value problems and understanding the real needs behind each project. I think strategically about long-term technical direction, balancing engineering trade-offs with business objectives to create scalable, sustainable solutions. As a problem-solver, I approach challenges analytically—identifying root causes, evaluating options, and developing practical, effective solutions. Across backend, frontend, and full-stack work, I bring a thoughtful, forward-looking approach that strengthens both the product and the team.

EXPERIENCE

Java Engineer, Amazon

06/2024 – Present
Arlington, Virginia, US

- **Real-Time Systems Orchestration:** Engineered and automated high-availability systems for global theme park operations, specifically focusing on the synchronization and real-time monitoring of attraction wait times and showtime scheduling for millions of annual visitors.
- **Event-Driven Microservices:** Designed a decoupled, high-throughput microservices ecosystem using Node.js and Java Spring Boot. Implemented Apache Kafka as the primary event bus to ensure reliable, sub-second data propagation between sensors, backend services, and guest-facing applications.
- **Core Infrastructure Innovation:** Developed a high-performance Redis Adapter library that was adopted as a core dependency across the enterprise platform. This standardized caching layer optimized data retrieval speeds and reduced primary database load by over 50%.
- **Scalable Service Design:** Built and deployed mission-critical microservices from the ground up using Node.js/Express.js, employing the Adapter Pattern to allow the system to switch between different data providers and protocols without breaking core business logic.
- **Performance Engineering & TDD:** Delivered production-grade, modular code under a strict Test-Driven Development (TDD) methodology. Achieved and maintained 100% test coverage for both Node.js and Spring Boot services, ensuring system stability during peak park attendance.
- **Modern Frontend Architecture:** Developed responsive, real-time dashboards for park operations using React and Angular. Created a library of reusable React Hooks and components specifically designed to visualize live streaming data from attraction sensors.
- **State Management & Data Fetching:** Optimized the frontend data layer using Redux and React Query. Implemented intelligent "stale-while-revalidate" caching strategies to ensure operators always had access to the most current park data with minimal API overhead.

- **UI/UX Performance Optimization:** Drastically improved dashboard responsiveness by refining rendering logic and reducing JavaScript bundle sizes. Partnered with UX designers to ensure all operational tools met strict accessibility standards while maintaining sub-second load times.
- **Technical Leadership & Troubleshooting:** Led cross-functional debugging efforts for high-priority incidents, maintaining 99.9% uptime for critical park systems. Conducted high-level code reviews and mentored junior developers on microservices best practices and clean architecture principles.

Senior Java Developer, Johnson & Johnson

10/2022 – 05/2024
New Brunswick, New
Jersey, US

- **Architected High-Scale Orchestration:** Designed and optimized mission-critical backend systems supporting Walmart's global e-commerce infrastructure. Spearheaded the Transaction-OL initiative, using Node.js and TypeScript to engineer high-throughput data contracts that power millions of Cart and Checkout sessions daily.
- **Native Android Development (Kotlin):** Developed and integrated native features within the Walmart Android app using Kotlin and Jetpack Compose. Bridged the gap between the Orchestration Layer and the mobile client, implementing MVI/MVVM architectures to ensure a responsive, state-driven user experience.
- **Mobile-First Performance Engineering:** Optimized the mobile data-consumption layer by implementing Kotlin Coroutines and Flow for asynchronous data handling. This reduced main-thread blocking and improved the "Time to Interactive" (TTI) for the checkout flow on low-end Android devices.
- **GraphQL & BFF Optimization:** Unified disparate microservices into a high-performance BFF (Backend for-Frontend) using GraphQL. Designed custom resolvers that allowed the Android native client to perform complex data fetches in a single round-trip, significantly reducing network latency and battery consumption.
- **Feature Leadership (Pet & Vet Experience):** Led the end-to-end implementation of the "Enhanced Pet Experience." Built the service logic in Node.js/Express.js and the corresponding UI components in Kotlin, creating a seamless, cross-platform journey for managing pet prescriptions and vet details.
- **Microservices & Spring Boot:** Engineered robust Spring Boot microservices to handle enterprise-level data requirements. Implemented advanced caching strategies and circuit breakers to ensure the Orchestration Layer remained resilient during high-traffic retail events (Black Friday/Cyber Monday).
- **Automated Quality & CI/CD:** Elevated the codebase to a perfect 100% test coverage using Jest and Cypress for web services, and JUnit/Espresso for Android native components. Ensured all code met strict SonarQube standards for security and maintainability.
- **Production Readiness & Stakeholder Alignment:** Actively collaborated with product owners and stakeholders to translate complex business requirements into production-ready features. Owned the lifecycle of prescription-related features, ensuring zero-downtime deployments and a bug-free user experience.

Java Developer, *United Airlines*

03/2020 – 09/2022
Chicago, Illinois, US

- **Multidisciplinary Engineering Leadership:** Spearheaded the development of scalable, enterprise-grade backend solutions for a diverse portfolio encompassing Healthcare, Fintech, and Real Estate. Orchestrated full-stack architectures using Java (Spring Boot) and Angular, aligning technical roadmaps with complex business logic to serve a rapidly expanding user base.
- **Legacy Modernization & JVM Optimization:** Led the strategic migration from Java 8 to Java 11, leveraging modularity and enhanced garbage collection to improve system throughput and security posture. This transition reduced technical debt and enabled the adoption of modern reactive programming patterns.
- **Polyglot Data Persistence:** Engineered a hybrid data layer optimizing for consistency and speed. Implemented PostgreSQL with Hibernate for complex relational transactions in Fintech, while utilizing AWS DynamoDB for high-velocity, schema-less data operations in e-commerce modules.
- **Cloud-Native Infrastructure (AWS & Kubernetes):** Architected and managed a fault-tolerant cloud ecosystem on AWS (EC2, S3, RDS, Lambda). Leveraged AWS EKS (Elastic Kubernetes Service) to containerize and orchestrate microservices, enabling auto-scaling capabilities that handled peak traffic surges with 99.99% availability.
- **Performance Engineering & Caching:** Drastically reduced API latency and server load by implementing a multi-tier caching strategy using Redis. This optimization improved frontend responsiveness by over 40%, directly enhancing user retention and system efficiency.
- **Advanced API Design (GraphQL & Python):** Designed flexible, high-performance APIs using GraphQL across Java and Python services. By implementing a unified data graph, empowered frontend teams to reduce network overhead and iterate faster on UI/UX requirements.
- **High-Volume Data Processing:** Developed robust, automated data pipelines using Spring Batch. Engineered complex cron-driven workflows to ingest and analyze multi-source third-party API data, ensuring data integrity for critical business reporting.
- **Frontend Excellence:** Crafted intuitive, responsive Single Page Applications (SPAs) using Angular, TypeScript, and Material UI. Focused on component reusability and state management to deliver a seamless, high-performance user interface across all device types.
- **Technical Mentorship & Quality Governance:** Directed rigorous code review cycles and established "Best Practice" documentation. Mentored junior engineers in clean code principles and SOLID design patterns, raising the overall engineering bar for the organization.

Junior Java Developer, *Berkshire Hathaway*

09/2018 – 02/2020
Omaha, Nebraska, US

- Developed and maintained scalable **RESTful APIs** using **Java 17** and the **Spring Boot** framework.
- Implemented robust **Microservices** architecture with **Spring Cloud** to improve system modularity and fault tolerance.
- Leveraged **Hibernate** and **Spring Data JPA** for efficient Object-Relational Mapping (ORM) and database management.
- Designed and optimized complex SQL queries and schemas for **PostgreSQL** and **MySQL** databases.
- Integrated **Spring Security** with **JWT** (JSON Web Tokens) to implement secure authentication and authorization protocols.
- Utilized **Maven** and **Gradle** for project dependency management and automated build processes.

- Wrote comprehensive unit and integration tests using **JUnit 5** and **Mockito** to ensure high code quality and coverage.
- Collaborated in an **Agile/Scrum** environment, utilizing **Git** for version control and **Jira** for task tracking.
- Streamlined deployment pipelines using **Docker** containerization and **CI/CD** tools like Jenkins or GitHub Actions.
- Monitored and debugged application performance issues using **Logback** and integrated messaging queues like **RabbitMQ** or **Kafka**.

EDUCATION

Bachelor's Degree in Computer Science, *University of Florida*

2014 – 2018

TECHNICAL SKILLS

Languages

Java, Kotlin, JavaScript, Ruby, Python

Communication

Effective communication, REST API, GraphQL.

Pattern

- Languages: Kotlin, Java
- Architecture: MVC, Clean Architecture
- Networking/DL: TCP/IP, Docker

Databases & Storage

MySQL, PostgreSQL, Redis, Amazon RDS, Google Cloud SQL

Testing & Tools

Tools: Git, Jenkins, JUnit, Selenium, Jira, Confluence, Agile/Scrum

Frameworks

Spring Boot/Backend, Spring Security, Hibernate, JSP, Express.js

Mobile Development

Frontend

- HTML5, CSS, JavaScript, TypeScript
- Frameworks: jQuery, React.js, Bootstrap, Angular 12, Angular 15, Vue
- Designs, Tailwind CSS, Material Design, Bootstrap 4

Cloud & DevOps

- AWS: EC2, S3, IAM, CloudFormation, Lambda, RDS, ElastiCache, CloudFront, CloudWatch, CloudTrail, CloudKinesis, CloudSearch, CloudFront, CloudFront, CloudFront
- Containerization: Docker, Kubernetes, Jenkins, GitLab CI, GitHub Actions, ArgoCD